Chapter 14, Problem 58P

Problem

The circuit parameters for a series *RLC* bandstop filter are $R = 250\ \Omega$, $L = 1$ mH, $C = 40$ pF. Calculate:

(a) the center frequency

(b) the half-power frequencies

(c) the quality factor

Step-by-step solution

Step 1 of 3

(a)

Calculate the center frequency of band-stop filter.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute $1\ \text{mH}$ for L and $40\ \text{pF}$ for C.

An *RLC* band stop filter has the following parameters $R = 250\ \Omega, L = 1\ \text{mH}, C = 40\ \text{pF}$

(a)

Centre frequency is given by

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{(10^{-3})(40\times10^{-12})}}\\ &= \frac{1}{2\times10^{-7}}\\ &= 5\ \text{Mrad/s}\end{aligned}$$

Therefore, the center frequency, ω_0 is $\boxed{5\ \text{Mrad/s}}$.

Step 2 of 3

(b)

Calculate the Bandwidth.

$$B = \frac{R}{L}$$
$$= \frac{250}{1 \times 10^{-3}}$$
$$= 250 \text{ krad/s}$$

Calculate the half power frequencies.

$$\omega_1 = \omega_0 - \frac{B}{2}$$
$$= 5000 \times 10^3 - \frac{250 \times 10^3}{2}$$
$$= 4.875 \text{ Mrad/s}$$

And,

$$\omega_2 = \omega_0 + \frac{B}{2}$$
$$= 5000 \times 10^3 + \frac{250 \times 10^3}{2}$$
$$= 5.125 \text{ Mrad/s}$$

Hence, the half power frequencies are $\boxed{4.875 \text{ Mrad/s, and } 5.125 \text{ Mrad/s}}$.

Step 3 of 3

(c)

Calculate the quality factor.

$$Q = \frac{\omega_0}{B}$$
$$= \frac{5 \times 10^6}{250 \times 10^3}$$
$$= 20$$

Hence, the quality factor, Q is $\boxed{20}$.